

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line (Base)

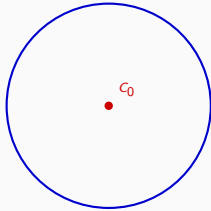
C_0 → Initial Center Point

AA' → Directing Line

Axis → Center Line



Step 1: Draw Generating Circle & Directing Line



$C_0 \rightarrow$ Initial Center Point

$AA' \rightarrow$ Directing Line

Axis $\rightarrow$ Center Line